

29. Morphological Operation – Erosion

Aim

To perform morphological erosion on an image using OpenCV.

Procedure

1. Read the image.
2. Convert it to grayscale.
3. Create a kernel.
4. Apply erosion using cv2.erode().
5. Display the original and eroded images.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

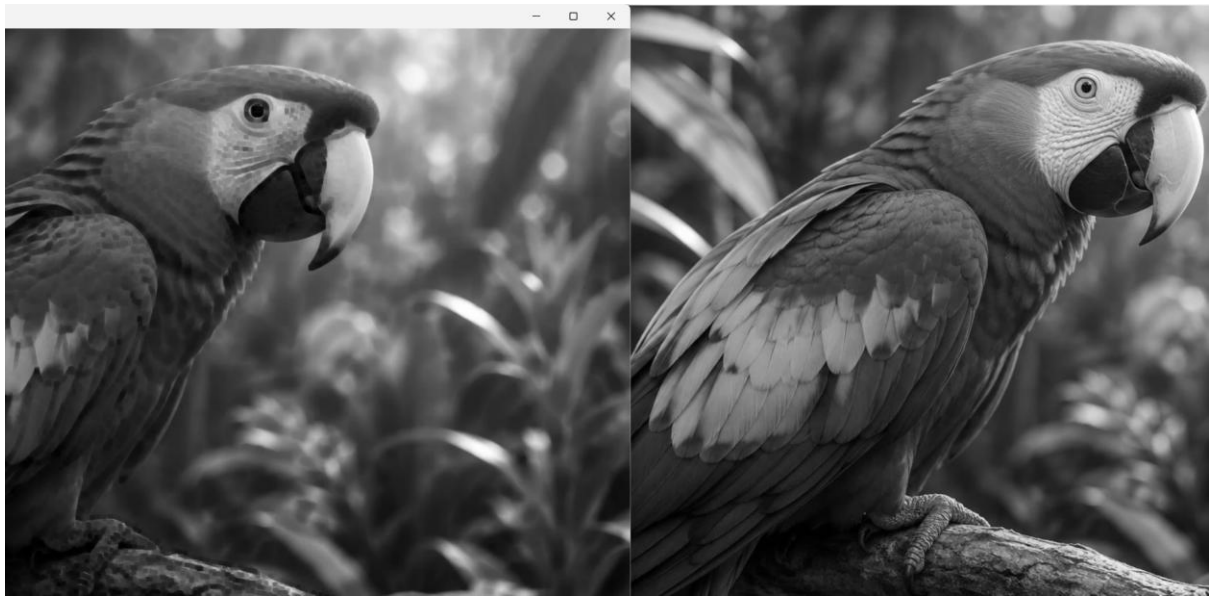
kernel = np.ones((5, 5), np.uint8)

erosion = cv2.erode(gray, kernel, iterations=1)

cv2.imshow("Original", gray)
cv2.imshow("Erosion", erosion)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

output



Result

The given image was successfully eroded using the morphological erosion technique.